

MOSAIC: auditable RNA–ADT annotation with branch-level modality evidence under donor shift and incomplete protein panels

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Abstract

Motivation: Fine-grained CITE-seq annotation under donor and protein-panel shift requires both a label and evidence for reviewing uncertain decisions.

Results: We present MOSAIC, an evidence-emitting RNA–ADT classifier with separate modality branches, margin-based arbitration, bounded sibling refinement and cell-level audit records. In nested leave-one-donor-out evaluation of 161,764 PBMCs and 58 L3 labels, MOSAIC reached M-F1 0.8072 [0.7917, 0.8227] (primary endpoint), W-F1 0.9108 [0.9062, 0.9153] and ACC 0.9119 [0.9086, 0.9153]. Versus a loss- and selection-matched early-fusion MLP, paired gains were +0.0111 M-F1 [0.0062, 0.0160] and +0.0074 balanced accuracy [0.0013, 0.0135], while ACC and W-F1 intervals crossed zero. A donor-held-out audit score combining final uncertainty and branch evidence improved error-detection AUROC from 0.6914 [0.6507, 0.7320] to 0.8401 [0.8267, 0.8536] and AUPRC from 0.3100 [0.2906, 0.3294] to 0.3587 [0.3478, 0.3696] against a 0.0881 held-out error rate; at 50% accepted coverage, error risk fell from 0.0565 to 0.0169, with no gain near 90% coverage. Branch evidence is therefore most useful for high-precision triage rather than near-full-coverage deployment. CLR improved both methods and reduced the M-F1 difference to +0.0064 [-0.0012, 0.0139], while masking and leave-class-out tests identified panel- and label-dependent failure modes.

Availability and implementation: Code, configurations, compact evidence tables, one P1/seed-42 checkpoint and a ten-cell CPU audit demo are available under the MIT License at <https://github.com/printfnice/MOSAIC> in the tagged submission snapshot; raw data, caches and full prediction files are not included, and no Zenodo DOI is claimed. The audited runs used Python 3.8.20, PyTorch 2.0.1+cu118 and CUDA on an NVIDIA RTX 4090.

Keywords single-cell annotation, CITE-seq, multimodal learning, donor shift, missing modality, selective prediction, model auditing

Introduction

Single-cell RNA sequencing and CITE-seq jointly measure transcriptomic programs and surface-protein markers at cellular resolution [18, 33, 41]. RNA captures broad lineage and cell-state programs, whereas antibody-derived tags (ADTs) directly measure canonical surface markers. Integrated references now resolve fine immune populations [7, 32, 39, 42], but supervised annotation remains difficult when labels separate closely related naive, central-memory, effector-memory, regulatory and activated states.

The practical value of an annotation model is determined by more than average accuracy on a convenient held-out cell set. Target cohorts often differ in donor composition, antibody panel, disease

state and missing cell states; a classifier may still output a confident fine label after the relevant sibling boundary has shifted or a defining protein marker is absent. A useful model should therefore report not only the fine label but also modality evidence, confidence and signals that the fine decision may need review.

Reference classifiers and semi-supervised systems include SingleR [4], CellTypist [11], scmap [23], scPred [3], scClassify [27], TOSICA [9] and scNym [22]. scVI [30], scANVI [45], totalVI [14], scvi-tools [13] and scArches [31] provide probabilistic representation learning, semi-supervised annotation and multimodal integration. Reference-mapping systems such as Symphony and Azimuth emphasize stable atlas transfer and multimodal integration [21, 19]. Batch-correction and atlas-integration methods address related distribution-shift problems [5, 17, 24]. Benchmark rankings vary with task, modality

and preprocessing [1, 28], and methods such as MMoCHi and STARCAT show that curated hierarchies or programs can be strong when the panel and label space match the task [8, 25]. These methods motivate evaluation that distinguishes predictive performance, robustness and biological interpretability rather than relying on a single headline score.

Existing methods cover many parts of this landscape, but prediction and evidence are usually treated as separate outputs. We address this gap by studying whether a multimodal annotation model can make its decision path inspectable while improving rare-label-sensitive donor generalization. Our central design principle is to treat each fine label as an evidence-emitting decision: RNA and ADT branches remain individually visible, fusion is driven by branch evidence, local refinement is bounded to declared sibling states, and the resulting record can be audited at cell level.

We introduce MOSAIC, a compact dual-modal classifier designed around this decision interface. Each prediction can be accompanied by per-modality labels and margins, the applied fusion policy, any bounded HSR action and ranked RNA and ADT features stored in checksum-indexed artifacts. RNA, ADT and fused representations receive separate supervision; a training-only teacher supplies soft targets; branch logits are combined using top-two margins; and an HSR head can modify only prespecified local labels. Attention and sparse-expert models motivate modular prediction paths [40, 44], whereas MOSAIC isolates the evidence interface from backbone scale with compact multilayer perceptrons.

We make four linked contributions: (i) an evidence-emitting dual-path architecture that preserves RNA, ADT and fused competence as inspectable prediction branches, with a +0.0111 donor-level M-F1 advantage over the matched MLP; (ii) a margin-based arbitration and audit interface that raises donor-held-out error-detection AUROC from 0.6914 to 0.8401; (iii) a bounded sibling-refinement path whose HSR action changes 0.14% of pooled predictions and remains enumerable for review; and (iv) a protocol-stratified failure-localization analysis spanning panel masking, leave-class-out rejection, attribution checks and an external hierarchy comparison.

Materials and methods

Datasets and donor-disjoint evaluation

The primary dataset is the paired PBMC 3P cohort from GSE164378 [18]: 161,764 cells from eight donors, 224 proteins and 58 fine-grained L3 labels. We used nested leave-one-donor-out evaluation (Supplementary Tables S1–S2). For test donor P1–P8, the validation donor was fixed as the next donor in a cycle, and the remaining six donors formed training. Cells were partitioned before preprocessing. The stored RNA and ADT layers were audited as non-negative, integer-like sparse CSC matrices with variable per-cell totals; we therefore treat the stored layers as raw-count-like inputs, while not claiming to reconstruct the full upstream download history (the unnumbered U0 preprocessing specification). The MOSAIC RNA path applied log1p once to this stored layer, selected 3,000 genes from training donors and standardized the selected matrix with a training-donor fit. For the locked primary ADT path, values were log1p-transformed, quantile-normalized to a normal output distribution using training donors, and then standardized using a second training-donor fit; the resulting transforms were applied unchanged to

validation and test cells. CLR sensitivity is reported separately using the centered log-ratio family of compositional transforms [2]; DSB normalization uses empty droplets and isotype/background controls [36], which are not available in this cohort. The eight held-out donors, not cells or model seeds, are the biological replicates for primary statistical comparison; fold sizes and the donor mapping are listed in Supplementary Table S1.

We used the L3 annotation level as the main endpoint because closely related immune states are most likely to be confused there. The L1 and L2 hierarchy was retained for parent fallback and interpretation, but not used to inflate the primary closed-set score. W-F1 reflects cohort-scale annotation quality, M-F1 reflects rare-label behavior and per-class F1 localizes sibling-level failures.

For a hierarchy-aware external comparator, we used the GSE229791 PDC101 sorted holdout with the official MMoCHi hierarchy and `external_holdout=True` [8]. Hash-tag oligonucleotide (HTO), isotype and control proteins were excluded because PDC101 sort labels are linked to HTO/source information. MOSAIC used the identical 2,098-cell holdout and the same processed RNA/ADT input scale; scalers were fitted on training cells only. MMoCHi used version 0.3.5 (commit `b0f3387`) and its native landmark-registration workflow where applicable. The comparison is therefore aligned at the holdout and label-hierarchy level while retaining package-specific transformations. It is reported separately from PBMC because the sorted PDC101 task has a different label space and uncertainty mode.

Secondary COVID-19, 5P and COMBAT/COVID-shift artifacts were retained only as breadth evidence, not pooled with the primary donor-level inference.

MOSAIC model architecture

For cell i , RNA and ADT inputs are $\mathbf{x}_i^{(r)}$ and $\mathbf{x}_i^{(p)}$. Modality encoders produce $\mathbf{h}_r = f_r(\mathbf{x}_i^{(r)})$ and $\mathbf{h}_p = f_p(\mathbf{x}_i^{(p)})$, and a fusion encoder maps their concatenation to $\mathbf{h}_f = f_f([\mathbf{h}_r; \mathbf{h}_p])$. The primary configuration uses batch-normalized MLP blocks of Linear–BatchNorm–ReLU–Dropout: two such blocks per modality encoder and three in the fusion encoder. The RNA and ADT paths are $3000 \rightarrow 512 \rightarrow 256$ and $224 \rightarrow 512 \rightarrow 256$, respectively, while the concatenated fusion path is $512 \rightarrow 512 \rightarrow 256 \rightarrow 256$. The final $256 \rightarrow 256$ projection defines the fixed fused interface used by arbitration and HSR. Each branch ends in a linear 58-class L3 head. The HSR delta head is linear and the gate head is a $314 \rightarrow 256 \rightarrow 1$ MLP with sigmoid output. Separate cross-entropy losses keep the modality branches predictive, so the model reports RNA-only, ADT-only and fused predictions from the same checkpoint.

The arbitration signal for branch b and cell i is its top-two logit margin $m_{b,i} = z_{b,i}^{(1)} - z_{b,i}^{(2)}$. Learned branch bias β_b and fixed temperature $\tau = 1$ define

$$w_{b,i} = \text{softmax}_b((m_{b,i} + \beta_b)/\tau), \quad \mathbf{z}_i^{\text{base}} = \sum_{b \in \{r,p,f\}} w_{b,i} \mathbf{z}_{b,i}. \quad (1)$$

The implementation learns only β_b ; the temperature is a fixed configuration value, and logits rather than probabilities are averaged. Raw margins are retained as branch evidence and interpreted as ordering signals rather than calibrated probabilities. The fixed temperature and learned bias are therefore part of the locked arbitration protocol, not post-hoc calibration parameters.

HSR receives the fused embedding and detached base probabilities. The two fixed sibling groups are CD4 TEM.3/CD4 TEM.4/CD4

MOSAIC: Evidence-emitting multimodal annotation

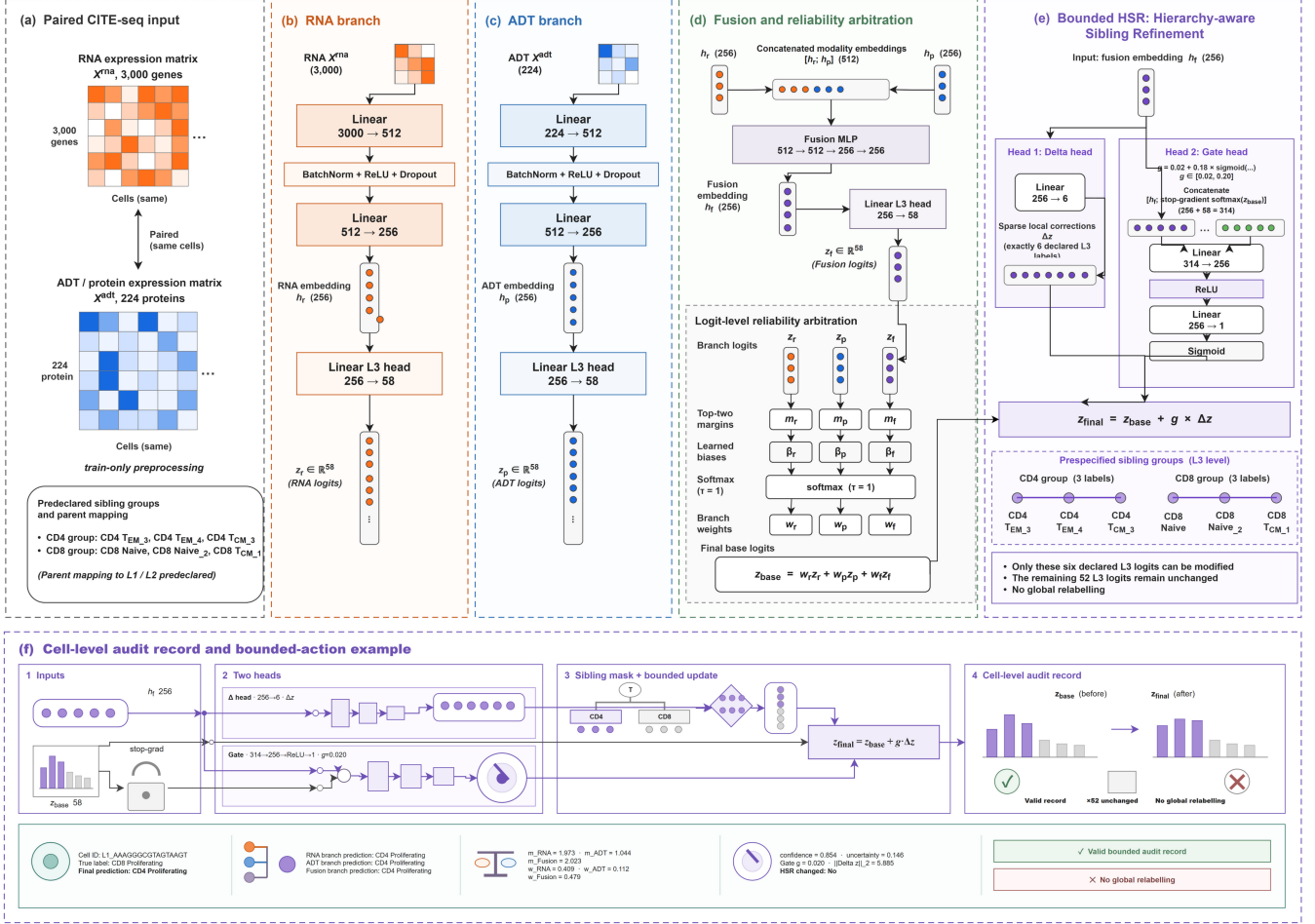


Figure 1 MOSAIC architecture. (a) Paired RNA ($N \times 3000$) and ADT ($N \times 224$) inputs with train-only preprocessing and prespecified sibling groups. (b) The RNA encoder maps $N \times 3000 \rightarrow N \times 512 \rightarrow N \times 256$ and feeds a 58-class L3 branch. (c) The ADT encoder maps $N \times 224 \rightarrow N \times 512 \rightarrow N \times 256$ and feeds a 58-class L3 branch. (d) Fusion maps the concatenated $N \times 512$ representation to $N \times 256$, followed by branch-margin arbitration and a 58-class fused output; the gate input is $256 + 58 = 314$. (e) Bounded HSR uses a six-output delta head scattered into the six declared L3 coordinates and a $314 \rightarrow 256 \rightarrow 1$ gate. (f) A cell-level audit record contains branch labels, margins, fusion information, the HSR action and feature-attribution fields.

TCM_3 and CD8 Naive/CD8 Naive.2/CD8 TCM_1; they were chosen from validation-donor confusion patterns before held-out evaluation. A six-output delta head predicts local corrections only within these groups. Let $S: \mathbb{R}^6 \rightarrow \mathbb{R}^{58}$ be the fixed scatter operator that places the six outputs at the declared L3 coordinates and fills the remaining 52 coordinates with zero. A bounded gate produces

$$g = 0.02 + 0.18 \sigma(\text{MLP}([h_f; \text{stopgrad}(\text{softmax}(z^{\text{base}})]))),$$

$$\Delta z_6 = h_\Delta(h_f),$$

$$z^{\text{final}} = z^{\text{base}} + g S(\Delta z_6).$$
(2)

All correction entries outside the six declared labels are set to zero by a fixed mask, so the remaining 52 L3 logits are not directly corrected. During training, $\mathcal{L}_{\text{HSR}} = \text{CE}(S(\Delta z_6), y \mid y \in M)$ is a 58-way cross-entropy evaluated only on cells whose true label is in the six-label mask M ; the scatter operator therefore keeps the global class coordinates while the loss is label-conditional. Base probabilities are detached for the HSR inputs, preventing the local auxiliary loss from changing the base branch logits through that path and avoiding

gradient pollution of the base classifier. The HSR head itself remains trainable end-to-end. This restricted loss is intentional: the six labels represented 12.35% of held-out cells, so λ_{HSR} is an auxiliary, label-conditional term rather than a uniform 58-class regularizer. The gate range $[0.02, 0.20]$ was fixed before the primary test pass: the small floor keeps the HSR path differentiable and records near-zero actions explicitly, while the no-HSR control provides the exact disabled-path comparison. Every activation is therefore enumerable and auditable, with the aggregate effect evaluated separately from the evidence emitted by the path.

Training and statistical analysis

Each donor fold used seeds 41, 42 and 43. The matched early-fusion MLP used the same RNA-ADT feature access, sqrt-balanced cross-entropy weighting, AdamW schedule, dropout, batch size and validation constraint-macro checkpoint-selection rule as MOSAIC; it differs in having no separate modality encoders, branch heads, margin arbitration or HSR. The primary comparator has 1.73M

parameters versus 2.51M for MOSAIC. We therefore added a near-capacity control with the same nested donor-seed protocol and matched training/selection settings but 704/256 hidden dimensions, yielding 2,467,706 parameters; its results are reported in Table 1 and Supplementary Table S2. The seed-42 MLP supplied stored training-cell probabilities to MOSAIC; validation and test teacher probabilities were never training inputs. Because the locked MOSAIC path distills from this matched MLP, the no-KD row provides a teacher-independent comparison. We retrained MLP, MOSAIC full, MOSAIC without HSR and MOSAIC without KD, and evaluated full checkpoints without retraining as RNA branch, ADT branch, fusion branch, uniform fusion, learned margin gate and learned margin gate plus HSR. The locked full path is the canonical report because it is the prespecified path that emits branch weights, local HSR actions and the complete audit record; no-KD, uniform-fusion and capacity-control rows provide performance controls. This design separates aggregate metric optimization from the value of an evidence-emitting interface.

The implemented objective combines sqrt-balanced cross-entropy, branch supervision, fixed-teacher probability distillation and the local HSR loss:

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{wCE}} + \lambda_b(\mathcal{L}_r^{\text{CE}} + \mathcal{L}_p^{\text{CE}}) + \lambda_f \mathcal{L}_f^{\text{CE}} \\ & + \lambda_{\text{KD}} T^2 D_{\text{KL}}(\mathbf{q} \parallel \mathbf{p}^T) + \lambda_{\text{HSR}} \mathcal{L}_{\text{HSR}}. \end{aligned} \quad (3)$$

For class c with training count n_c , the sqrt-balanced weight in $\mathcal{L}_{\text{wCE}}$ is proportional to $1/\sqrt{n_c}$ and normalized to mean one, so $\mathcal{L}_{\text{wCE}} = -\sum_i \omega_{y_i} \log p_{i,y_i}$; this choice explicitly addresses the rare-label imbalance described in prior systematic evaluations [6]. In the distillation term, the stored teacher vector $\mathbf{q}$ is clipped to the supported positive range and row-normalized before the KL calculation, while $\mathbf{p}^T = \text{softmax}(\mathbf{z}^{\text{final}}/T)$ uses the final student logits, including the HSR path. Thus KD can train the local delta and gate, whereas the no-KD run removes the teacher term while retaining the same architecture and HSR loss. The implementation does not apply a second temperature transform to the stored teacher probabilities. The factor T^2 is retained as a fixed loss-scale convention. Models used AdamW for at most 35 epochs, batch size 1,024, learning rate 10^{-3} , weight decay 10^{-4} , dropout 0.2 and patience 14. We set $\lambda_b = 0.35$, $\lambda_f = 0.50$, $\lambda_{\text{KD}} = 0.05$, $T = 2$ and $\lambda_{\text{HSR}} = 0.20$ before test evaluation. MOSAIC checkpoints maximized validation M-F1 subject to validation ACC and W-F1 being within 0.0015 of their best observed values.

We report ACC, W-F1, M-F1, balanced accuracy and per-class F1. M-F1 was treated as the primary aggregate endpoint because it is most sensitive to rare sibling labels. The main M-F1 is a run-level aggregate; a supplementary table recomputes support-aware per-class macro F1 after zero-support exclusion, so its small numerical difference is an aggregation convention rather than a second endpoint. Macro recall in that decomposition is the same support-aware balanced-accuracy calculation. For rejection, combined hierarchical accuracy is $[n_{\text{accepted correct L3}} + n_{\text{rejected correct parent}}]/n_{\text{all}}$. Top-label conflict is $\mathcal{K}[\{\hat{y}_r, \hat{y}_p, \hat{y}_f\}] > 1$, and deletion lift is the target-probability decrease after deleting the top 20 attribution-ranked features minus the matched random-feature decrease. Unless stated otherwise, intervals are two-sided 95% Student- t intervals across eight held-out donors after averaging three seeds within each donor. Seeds were averaged within each held-out donor; donor means, paired-difference intervals and Wilcoxon sensitivity tests

were then calculated across eight donors. The support-aware macro-F1 decomposition has paired $P = 0.00105$ before any multiplicity adjustment, but is interpreted as a mechanism analysis rather than an additional primary endpoint. No cell-level bootstrap was used for the primary PBMC comparison, and exploratory attribution enrichment tests are not used as family-wise confirmatory claims.

Robustness, rejection and interpretability analysis

Each of the 24 full checkpoints was frozen and evaluated with full input; deterministic masks removing 10, 30, 50 or 70% of proteins; prespecified memory-marker and T-cell-marker masks; and RNA-only input. Masks were fixed before test evaluation. These are stress tests of a full-input model under panel mismatch, not mask-aware training experiments. For a separate calibration audit, one positive scalar temperature was fitted by validation negative log likelihood for each donor-seed checkpoint and then applied once to the untouched test probabilities. Temperature selection did not use test labels and was not used to alter the locked primary predictions.

Because every natural donor fold retained all 58 labels in training, unknown-label evidence used five prespecified leave-class-out targets: gdT.2, NK.3, CD4 TCM.1, CD8 TEM.4 and B naive lambda. Each model was retrained on 57 labels using one fixed seed-42 split; three model seeds were evaluated per target, so the targets and seed replicates, not eight donors, define this stress-test unit. Thresholds for one minus maximum probability, one minus top-two margin and energy were selected using known validation cells to target 0.95 or 0.80 known coverage. Energy was $E(\mathbf{z}) = -\log \sum_c \exp(z_c)$ at temperature $T = 1$, following energy-based OOD scoring [29]. Known coverage is the fraction of known-label cells accepted for L3 output; unknown recall is the fraction of removed-label cells rejected; parent-safe rate is the fraction of rejected unknown cells whose predicted parent label is correct; and combined hierarchical accuracy counts accepted cells by L3 correctness and rejected cells by parent correctness. These scores are related to confidence calibration, uncertainty estimation and selective prediction [12, 15, 16, 20, 26].

Gradient-times-input attribution was computed for up to 64 correctly predicted cells per donor, seed and sibling label, with RNA and ADT summarized separately. We quantified donor-pair Spearman correlation, top-20 Jaccard overlap and canonical-marker enrichment using prespecified class-modality marker sets and 5,000 permutations. Deletion lift is the reduction in target probability after deleting top-ranked attribution features minus the matched reduction after deleting random features. Integrated gradients checked directional agreement on a small subset [43]; these checks are complementary to broader feature-attribution and explanation frameworks [34, 35, 37, 38].

To extend the audit beyond the selected sibling classes, we separately evaluated all 58 labels on the 24 current full checkpoints. A representative marker-family map was used to count panel availability and top-20 gradient-times-input hits among up to eight correctly predicted cells per label, checkpoint and modality. This map is an evidence vocabulary rather than a complete marker ontology; the resulting coverage and CD8/PDC101 linkage are descriptive and are not interpreted as causal biomarker evidence.

For the combined audit score, each target donor k was evaluated only with out-of-fold cell-level predictions from its own held-out fold; the logistic model was fitted on the seven corresponding held-out-donor folds $j \neq k$ and then applied to donor k . Thus, no in-sample

predictions from the training cells of the target fold entered audit-score fitting.

Software implementation and reproducibility

The Python 3.8.20/PyTorch 2.0.1+cu118 implementation saves configuration, split manifest, selected checkpoint, predictions, per-class metrics, run log and SHA-256 artifact index for every training unit. Table and figure scripts consume declared CSV/JSON artifacts and reject incomplete matrices. The repository also contains historical TOSICA-derived prototypes for provenance, but the reported MOSAIC experiments use the scripts and artifacts declared in the MOSAIC evidence registry rather than the TOSICA pathway-transformer training path [9]. Experiments used one NVIDIA RTX 4090. MOSAIC full had 2.51M trainable parameters and a median runtime of 130.0 seconds per fold-seed unit; the 1.73M matched-protocol MLP had a median runtime of 75.8 seconds, and the 2.47M capacity-control MLP had a median runtime of 80.7 seconds.

Results

Donor-disjoint annotation performance

Across eight held-out donors, MOSAIC full reached ACC 0.9119 [0.9086, 0.9153], W-F1 0.9108 [0.9062, 0.9153] and M-F1 0.8072 [0.7917, 0.8227] (Table 1; Supplementary Table S2; Fig. 2a). Relative to the matched MLP protocol, paired donor differences were +0.0016 ACC [−0.0011, +0.0044], +0.0016 W-F1 [−0.0009, +0.0041], +0.0111 M-F1 [+0.0062, +0.0160] and +0.0074 balanced accuracy [+0.0013, +0.0135]. Five of eight donors improved in ACC, four in W-F1, all eight in M-F1 and seven in balanced accuracy. After matching the sqrt-balanced loss and checkpoint rule, the remaining macro advantage was predominantly a precision effect rather than a recall effect: macro precision increased by +0.0171 [+0.0111, +0.0230], whereas macro recall increased by +0.0074 [+0.0013, +0.0135]. MOSAIC therefore made fewer spurious assignments into rare sibling labels rather than simply recovering more of them; this is consistent with margin-based arbitration, which down-weights a branch with a small top-two margin. The paired Cohen d_z values for ACC, W-F1, M-F1 and balanced accuracy were 0.498, 0.544, 1.892 and 1.015, respectively. ACC and W-F1 remained comparable under the donor-level intervals. Branch-sensitivity results are detailed in Supplementary Table S4.

The paired t -test gave $P = 0.0239$ for balanced accuracy, whereas the Wilcoxon sensitivity test gave $P = 0.0547$; with eight donor units we report both and treat this gain as marginal. For the four prespecified aggregate comparisons, Bonferroni adjustment leaves the M-F1 result below 0.005 ($P = 0.0011$), while the other endpoints remain non-confirmatory.

We also reran one published single-cell annotation package baseline and nine classical or tree-based baselines using all available training donor cells in each fold. With the legacy per-class training cap removed, early-fusion XGBoost reached ACC 0.9032, W-F1 0.9010 and M-F1 0.7885. MOSAIC full retained higher point estimates and a higher worst-donor ACC than this uncapped comparator (0.9057 versus 0.8750). Paired donor differences were +0.0087 ACC [−0.0029, +0.0204], +0.0097 W-F1 [−0.0010, +0.0205], +0.0187 M-F1 [+0.0053, +0.0321] and +0.0330 balanced accuracy [+0.0190, +0.0469]; Cohen d_z values were 0.629, 0.755, 1.167 and 1.977, respectively. XGBoost used 80 trees, depth 4, learning rate 0.08,

subsample 0.9, column subsample 0.8 and the histogram CUDA training path; prediction/evaluation used materialized CPU arrays. Full protocol details and source configurations are provided in the Supplementary Material.

The capacity control reached ACC/W-F1/M-F1/balanced accuracy 0.9111/0.9096/0.8001/0.8024. MOSAIC minus this control was +0.0008 ACC [−0.0017, +0.0033], +0.0011 W-F1 [−0.0011, +0.0033], +0.0071 M-F1 [+0.0031, +0.0111] and +0.0060 balanced accuracy [+0.0001, +0.0120] across the eight donor means. The near-capacity control therefore narrows, but does not remove, the attribution gap; the evidence supports a joint representation-and-audit advantage rather than a strict single-module causal claim.

The full-transcriptome-denominator CellTypist-compatible RNA-only comparison reached ACC/W-F1/M-F1/balanced accuracy 0.8534/0.8497/0.6835/0.6933 under the primary donor-disjoint protocol. This memory-safe runner calls the CellTypist train/annotate API after applying full-transcriptome CP10K and log1p, then retains the train-only 3,000-gene subset; it is reported as package-compatible rather than fully native because the retained matrix no longer passes the package's sum-to-10,000 expression check. The older subset-first row is retained only as a sensitivity and is lower for M-F1 but higher for ACC/W-F1. Published latent baselines are reported separately on a fixed strict split: scANVI reached ACC/W-F1/M-F1 0.8907/0.8882/0.7729 and totalVI 0.8100/0.8251/0.6916. On the independent E-MTAB-10026 patient-disjoint cohort, MOSAIC reached ACC/W-F1/M-F1 0.8590/0.8550/0.6471 versus 0.8553/0.8514/0.6411 for an early-fusion MLP. This second cohort reproduces the direction of the point-estimate advantage under a distinct patient-disjoint protocol; its three-seed M-F1 interval crossed zero and the seeds are not patient-level biological replicates. The protocol-stratified matrix is provided as descriptive cross-dataset breadth rather than a pooled ranking.

The 48-run CLR sensitivity matrix (24 fold-seed units for each method) shows that ADT preprocessing is itself a substantial source of performance variation. MOSAIC increased by 0.0031/0.0029/0.0104/0.0116 for ACC/W-F1/M-F1/balanced accuracy, while the matched MLP increased by 0.0031/0.0035/0.0151/0.0164. Under CLR, MOSAIC minus MLP was +0.0016 ACC [−0.0006, +0.0038], +0.0009 W-F1 [−0.0016, +0.0035], +0.0064 M-F1 [−0.0012, +0.0139] (paired $P = 0.0861$) and +0.0025 balanced accuracy [−0.0039, +0.0089]. The locked quantile-normalized transform remains the prespecified primary protocol; the sensitivity result shows that preprocessing and architecture operate on comparable scales and should be interpreted jointly.

We also evaluated a full-transcriptome RNA library-size normalization sensitivity across 48 retraining units (24 fold-seed units per method; Supplementary Table S24). Full-transcriptome CP10K followed by log1p produced MOSAIC ACC/W-F1/M-F1 of 0.9122/0.9107/0.8088 and matched-MLP values of 0.9094/0.9083/0.7949. The paired MOSAIC-MLP differences under this sensitivity were +0.0029 ACC [+0.0002, +0.0056], +0.0024 W-F1 [−0.0001, +0.0050] and +0.0139 M-F1 [+0.0059, +0.0220], compared with +0.0016/+0.0016/+0.0111 for the locked log1p protocol. Thus, library-size normalization changes absolute scores and the comparator gap, but does not reverse the direction of the macro comparison; the locked source-count path remains the

Table 1 Primary nested leave-one-donor-out PBMC L3 performance. Values are donor means with two-sided 95% Student-*t* intervals after averaging three seeds within each held-out donor; worst and best columns are point estimates across the eight held-out donors. Parameter counts are trainable parameters for neural models; XGBoost and CellTypist entries are marked not applicable. The two MLP rows use the matched objective and selection protocol; the second is a 2,467,706-parameter capacity control. The primary CellTypist-compatible row applies the full-transcriptome CP10K denominator before retaining train-only selected genes; the subset-first row is retained as a preprocessing sensitivity.

Method	Params	ACC	W-F1	M-F1	B-ACC	Worst ACC	Best ACC
MLP matched protocol	1.73M	0.9103 [0.9065, 0.9141]	0.9092 [0.9046, 0.9137]	0.7961 [0.7786, 0.8136]	0.8011 [0.7821, 0.8201]	0.9036	0.9153
MLP parameter-matched capacity	2.47M	0.9111 [0.9081, 0.9142]	0.9096 [0.9051, 0.9142]	0.8001 [0.7846, 0.8156]	0.8024 [0.7852, 0.8196]	0.9059	0.9166
MOSAIC full	2.51M	0.9119 [0.9086, 0.9153]	0.9108 [0.9062, 0.9153]	0.8072 [0.7917, 0.8227]	0.8085 [0.7890, 0.8280]	0.9057	0.9183
MOSAIC without HSR	2.42M	0.9135 [0.9102, 0.9168]	0.9121 [0.9079, 0.9162]	0.8097 [0.7950, 0.8243]	0.8067 [0.7895, 0.8238]	0.9081	0.9200
MOSAIC without KD	2.51M	0.9134 [0.9102, 0.9167]	0.9121 [0.9078, 0.9164]	0.8110 [0.7956, 0.8265]	0.8079 [0.7913, 0.8244]	0.9092	0.9200
Early-fusion XGBoost (uncapped)	n/a	0.9032 [0.8912, 0.9152]	0.9010 [0.8901, 0.9120]	0.7885 [0.7648, 0.8122]	0.7755 [0.7543, 0.7968]	0.8750	0.9201
CellTypist-compatible full-denominator	n/a	0.8534 [0.8391, 0.8677]	0.8497 [0.8350, 0.8644]	0.6835 [0.6608, 0.7062]	0.6933 [0.6731, 0.7136]	0.8310	0.8712
CellTypist-compatible subset-first sensitivity	n/a	0.8652 [0.8573, 0.8732]	0.8565 [0.8477, 0.8654]	0.6559 [0.6249, 0.6870]	0.6240 [0.5950, 0.6530]	0.8486	0.8767

prespecified primary protocol and the CP10K result is a sensitivity, not a second primary endpoint.

Component trade-offs and auditability

Direct ablations separate prediction utility from evidence utility. Relative to the matched MLP, the no-KD path increased ACC by +0.0031 [+0.0003, +0.0059], W-F1 by +0.0030 [+0.0004, +0.0056], M-F1 by +0.0150 [+0.0091, +0.0208] and balanced accuracy by +0.0068 [+0.0003, +0.0133], with positive M-F1 differences for all eight donors (Supplementary Table S3). The no-HSR and no-KD paths reached M-F1 0.8097 and 0.8110, respectively, versus 0.8072 for the locked path; among these direct ablations, the locked path retained the highest balanced accuracy (0.8085 versus 0.8067 and 0.8079) while emitting the complete audit record. Within the same full checkpoints, uniform fusion exceeded learned margin gate plus HSR by 0.0011 ACC, 0.0010 W-F1, 0.0042 M-F1 and 0.0044 balanced accuracy. A branch-loss sensitivity matrix showed that $\lambda_b = 0.70$ gave the highest ACC/W-F1 (0.9140/0.9126), whereas fusion-supervision-only training gave the highest M-F1/balanced accuracy (0.8122/0.8132; Fig. 2b; Supplementary Table S4). The margin gate is therefore retained for its inspectable branch-weight signal, not because it is the aggregate-metric optimum under the locked unnormalized-margin implementation.

The dual-path macro benefit is already present without branch-level supervision: the $\lambda_b = 0$ model improves M-F1 by +0.0123 and balanced accuracy by +0.0086 over the matched early-fusion MLP. Branch supervision adds no aggregate performance in this sensitivity family; it converts the same architecture into an interface that reports per-modality labels and margins, at a measured cost of 0.0012 M-F1 and 0.0012 balanced accuracy relative to the $\lambda_b = 0$ configuration. The locked full configuration is the prespecified evidence-emitting operating point, selected to retain the complete branch-weight, local-action and audit-record outputs rather than to maximize every aggregate metric independently. The no-HSR and no-KD controls exceed the locked configuration on three of four aggregate metrics, whereas the locked configuration retains the complete audit record; these are distinct operating choices rather than a hidden claim of universal dominance.

A train-fold standardization sensitivity removed the scale difference between branch margins but did not change the component interpretation: raw, train-scaled and train-scaled-plus-HSR gates remained within the descriptive donor-level band, whereas uniform fusion was higher by +0.0037 M-F1 and +0.0033 balanced accuracy than the raw gate. The learned branch biases were -0.035 , -0.501 and $+0.632$ for RNA, ADT and fusion, against mean training margins of 3.782, 2.697 and 4.651, respectively.

Branch evidence carries information beyond final confidence

To quantify the value of the evidence interface, we tested whether branch signals predict errors (Fig. 2c). Three-branch top-label conflict achieved error-detection AUROC 0.7552 [0.7445, 0.7659], compared with 0.6914 [0.6507, 0.7320] for final uncertainty; throughout, bracketed intervals denote two-sided 95% Student-*t* intervals unless stated otherwise. Continuous branch scores alone did not improve AUPRC or risk at 90% coverage, but the donor-held-out logistic audit score combining final uncertainty and branch evidence increased AUROC to 0.8401 [0.8267, 0.8536] and AUPRC to 0.3587 [0.3478, 0.3696]. The paired gains over final uncertainty were +0.1488 AUROC [0.1162, 0.1814] ($P = 1.29 \times 10^{-5}$) and +0.0487 AUPRC [0.0295, 0.0679] ($P = 5.47 \times 10^{-4}$). At 50% coverage, the combined score reduced accepted-cell risk from 0.0565 to 0.0169, a 3.3-fold reduction; at 90% coverage the change was +0.0002 ([-0.0022 , 0.0025], $P = 0.871$). The final-uncertainty curve is nearly flat between 50% and 90% coverage, which explains why branch evidence is useful for high-precision triage but not near-full-coverage selection. The pooled held-out error prevalence was 0.0881, and the six declared HSR labels represented 12.35% of held-out cells. Branch margins also ordered branch correctness (mean Spearman 0.3777; Fig. 2d). Across the 24 full fold-seed artifacts, HSR changed 688 of 485,292 pooled held-out predictions (0.14% overall; 1.15% among the six declared HSR labels), and same-checkpoint HSR effects were below the prespecified descriptive audit threshold of 0.005 for all four aggregate metrics (Supplementary Tables S5–S7). These results establish branch evidence as an additional error-audit channel that complements, rather than replaces, final uncertainty.

The corrected gate-bound audit gives prediction-change rates of 0.034%–0.149% across the [0,0.10], [0,0.50] and locked [0.02,0.20] ranges; all aggregate differences remained inside the descriptive 0.005 band. We use 0.005 only as a scale for reporting local perturbation: it is below the primary M-F1 paired standard deviation (0.0059) and close to the balanced-accuracy paired standard deviation (0.0073), not as an equivalence or non-inferiority margin. Table 2 pairs a positive agreement record with a modality conflict and a correct-to-wrong HSR change, making clear that the record supports review rather than automatic correctness.

Where proteins are missing matters more than how many

At 50% random protein masking, W-F1 was 0.8971, a donor-level decrease of 0.0137 from full input; at 70% masking it was 0.8921 (Fig. 2e). RNA-only input reached W-F1 0.8882, and M-F1 was more sensitive, decreasing by 0.0330 at 50% masking and 0.0514 for RNA-only input. Prespecified memory-marker and T-cell-marker masks reached W-F1 0.9082 and 0.9021, with M-F1 0.8006 and 0.7859, respectively (Supplementary Table S8). Removing the 15 T-cell ADTs (6.7% of the 224-protein panel) cost 0.0213 M-F1 versus 0.0168 for removing 30% of proteins at random (0.7859 versus 0.7904); normalized per removed-panel fraction, the targeted loss is approximately 5.7-fold larger. Panel composition therefore matters more than missingness fraction alone for these sibling states, identifying marker availability as a concrete deployment determinant. A separate training-only 30% ADT feature-dropout matrix improved M-F1 under the targeted T-cell mask from 0.7859 to 0.7884, but reduced full-input M-F1 from 0.8072 to 0.8016; the complete comparison is in Supplementary Table S23. This bounded intervention therefore provides a panel-aware sensitivity result rather than evidence that arbitrary missing panels are solved. In the validation-only temperature audit, test ECE decreased from 0.0713 [0.0645, 0.0781] to 0.0391 [0.0303, 0.0478], NLL from 0.4255 [0.3900, 0.4610] to 0.3904 [0.3432, 0.4376] and Brier score from 0.1523 [0.1456, 0.1591] to 0.1462 [0.1380, 0.1545], with no test argmax changes. This is a post-hoc calibration result, not evidence that panel masking is solved.

Rejection is bounded by marker availability

At target known coverage 0.80, one-minus-margin rejection achieved observed known coverage 0.7878, mean unknown recall 0.4032 and combined hierarchical accuracy 0.9146 (Supplementary Tables S9–S11). These estimates use one fixed seed-42 leave-class-out split for each of five prespecified target labels and three model seeds per target; the target labels, not the eight donors, are the statistical units for this stress test. Unknown detection varied sharply by target: B naive lambda had AUROC 0.4258 and recall 0.1285, whereas other sibling targets were easier to rank. This pattern is mechanistically interpretable because IGKC, IGLC2 and IGLC3 were retained in the RNA feature set but the ADT panel had no kappa/lambda surface marker in any donor fold (Supplementary Table S12). Parent-safe rate among rejected unknown cells was 0.2259, so about 9.1% of unknown cells were both rejected and assigned to the correct parent at the default operating point (Supplementary Table S11). The result identifies a marker-availability boundary and supports manual review rather than automatic parent fallback. A separate current-checkpoint audit covers all 58 labels with representative panel availability and top-attribution hits; the five leave-class-out rows are descriptive and do not establish a general marker–rejection law (Supplementary unnumbered diagnostic).

Curated hierarchies win where the panel carries the boundary

On the PDC101 holdout, MOSAIC achieved 0.9077/0.9068/0.9007 ACC/W-F1/M-F1 versus 0.9347/0.9335/0.9298 for official MMoCHI (Supplementary Tables S13–S14). The largest gaps were concentrated

Table 2 Audit-record schema and concrete case records supported by the current MOSAIC artifacts. The schema lists field groups, provenance and local artifact status; the case block contains a positive three-branch agreement, a modality conflict and a sparse HSR correct-to-wrong change. These records illustrate structured review without implying that agreement or every local action guarantees the final label.

Field group	Stored fields	Source	Example value	Public artifact (Y/N)
Cell identity	cell.id, true label, final prediction	prediction/case-study CSV	L1_AAAGGGCGTAGTAAGT; CD8 Proliferating → CD4 Proliferating	Y (tagged package)
Modality evidence	RNA/ADT/fusion predictions and top-two margins	model audit record	CD4 Proliferating; CD4 Proliferating; CD4 Proliferating	Y (tagged package)
Arbitration evidence	branch weights and margins	model audit record	$w = (0.409, 0.112, 0.479)$	Y (tagged package)
Confidence	final confidence, uncertainty, conflict	model audit record	0.854; conflict=0	Y (tagged package)
HSR evidence	base/final action, gate, delta norm, changed flag	HSR audit record	$g = 0.020$; $\ \Delta\ _2 = 5.885$; changed=0	Y (tagged package)
Feature evidence	top-5 RNA and ADT attribution features	gradient-times-input audit	RNA: SPC24; EXO1; SPC25; GINS2; STMN1; ADT: CD278; CD28; CD71; CD38-2; CD4-1	Y (tagged package)

Case	Donor / seed / cell	Label and branch evidence	HSR/base-final evidence	Audit interpretation	Public artifact (Y/N)
Agreement correct	P8 / 42 / E2L6_GTCACGGTCCTGAAG	true/final=Eryth; RNA=ADT=fusion=Eryth; disagreement=0	all branches agree; no HSR change recorded	branch consensus is a positive audit record, not a correctness guarantee	Y
Modality conflict	P4 / 41 / L4_TTGTTGGACACGCTCA	Treg Naive → CD4 Naive; RNA=CD4 TCM.1, ADT=Treg Naive, fusion=CD4 Naive; disagreement=0.667	final uncertainty=0.640; outside declared HSR group	ADT and RNA disagree; review signal is present but the final label is wrong	Y
HSR changed correct → wrong	P1 / 41 / L1_CAAATGACAGTCTCTGA	true/base=CD4 TCM.1; final=CD4 TCM.3	$g = 0.0204$; $\ \Delta\ _2 = 12.940$; changed=1	bounded local action is sparse but not uniformly beneficial	Y

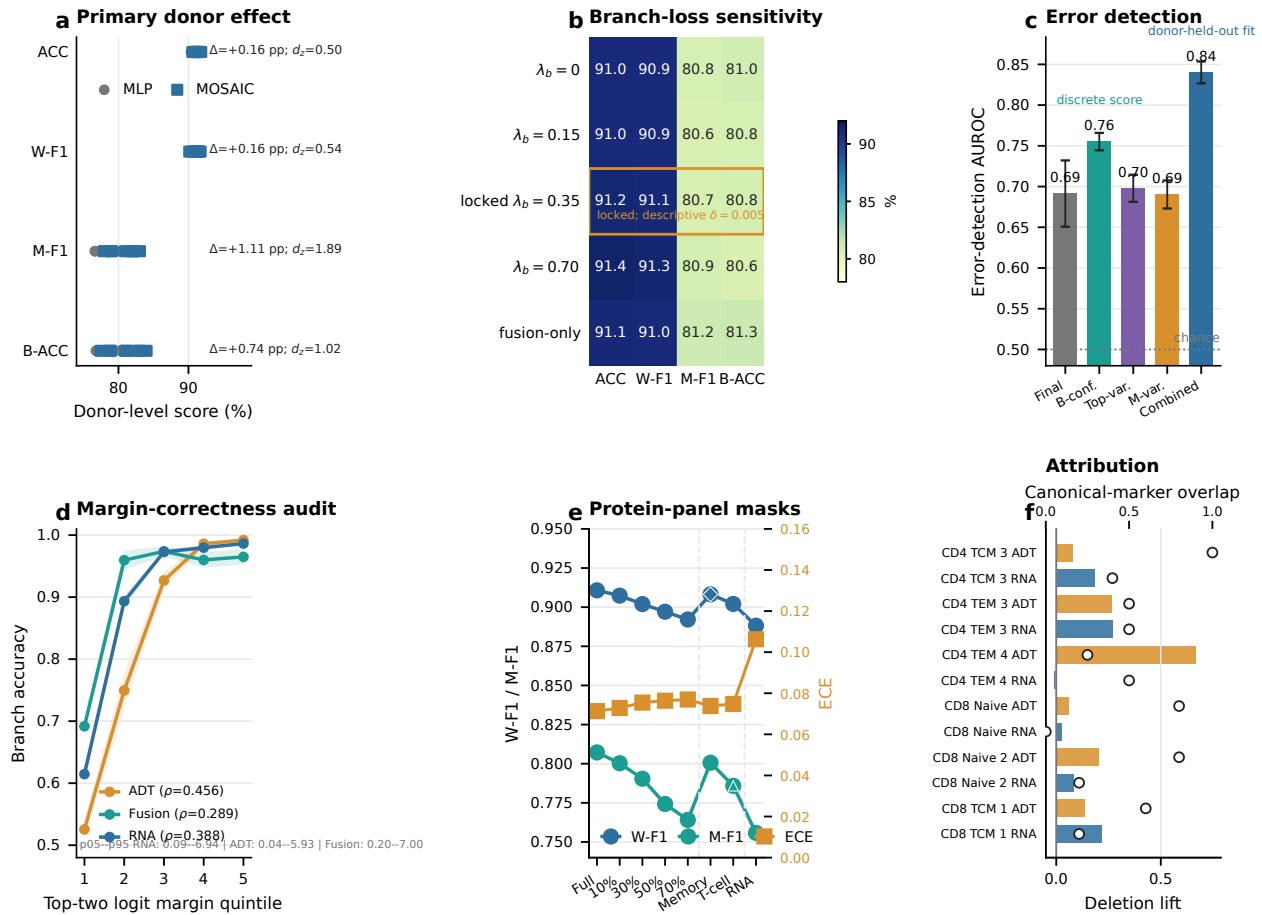


Figure 2 Donor-disjoint performance and mechanism evidence. All donor intervals use eight held-out donors after seed averaging. (a) Absolute donor-paired scores for MOSAIC full and the matched MLP, connected within donor, with mean intervals and Cohen's d_z annotations. (b) Retrained branch-loss sensitivity under the primary nested donor protocol; the boxed row is the locked configuration and the 0.005 line is descriptive only. (c) Error-detection AUROC for final uncertainty, branch-derived audit scores and the donor-held-out combined logistic score (0.8401); the pooled error prevalence is 0.0881. B-conf. denotes branch conflict, Top-var. top-probability variance and M-var. margin variance. (d) Branch accuracy across top-two logit-margin quintiles for RNA, ADT and fusion branches, with donor-level Spearman correlations and pooled margin ranges. (e) W-F1 and M-F1 use the left axis, while ECE uses the right axis under deterministic protein-panel masks; removing the 15 T-cell ADTs (6.7% of the panel) costs more M-F1 per removed feature than removing 30% at random. (f) Attribution deletion lift uses the left axis and canonical-marker overlap uses the separate top axis.

in CD8 memory boundaries (0.063, 0.061, 0.047 and 0.030 F1), whereas CD4 gaps were smaller and monocyte F1 was tied. The CD8 rows in the marker audit show that representative ADT evidence is present in both tasks, so the gap is reported as a boundary-localization comparison rather than attributed to marker absence or causal mediation. This contrast supports complementary operating regimes: curated marker hierarchies are strong in a small sorted label space, while MOSAIC exposes branch evidence and panel-stress behavior in reference-trained annotation.

Runtime, comparator, preprocessing, attribution, panel-aware and breadth protocols are enumerated in Supplementary Tables S15–S24, with source aggregation metadata retained for every secondary result.

Across donor pairs, attribution Spearman correlation ranged from 0.529–0.796 for RNA and 0.295–0.612 for ADT; 9/12 canonical-marker tests survived BH-FDR, and top-feature deletion exceeded matched random deletion in all 12 tests (mean lifts 0.1304 RNA and 0.2358 ADT; Fig. 2f; Supplementary Table S20). Two deliberately scoped counterexamples clarify the interpretation: CD4 TEM.4 ADT had a large deletion lift (0.6667) but low cross-donor stability

(Spearman 0.2945) and marker overlap 0.25 ($q = 0.3471$), whereas CD8 Naive RNA was highly stable (0.7956) but had zero canonical-marker overlap ($q = 1.0000$). Attribution is therefore functional evidence for the model's decision, not a claim that every stable feature is a canonical biomarker. These are reproducibility and faithfulness checks for selected sibling classes, not a family-wise biomarker study.

Discussion

The central result is an evidence-emitting decision path for fine-grained multimodal annotation under donor shift. In the nested donor-disjoint PBMC evaluation, MOSAIC improved M-F1 by +0.0111 and balanced accuracy by +0.0074 over the loss- and selection-matched MLP; the macro gain was driven mainly by precision and supported by recall. Its audit channel independently improved error-detection AUROC from 0.6914 to 0.8401 and AUPRC from 0.3100 to 0.3587, reducing risk at 50% coverage from 0.0565 to 0.0169 while providing no gain near 90% coverage. CLR changed both

Cell-level audit records

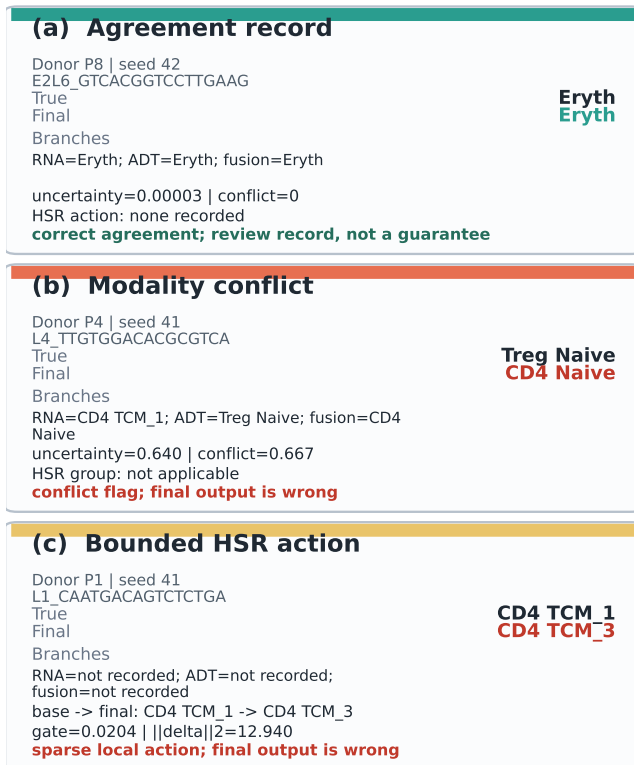


Figure 3 Cell-level audit records. Three checksum-linked records illustrate (a) correct branch agreement, (b) a modality conflict that flags a wrong final label and (c) a sparse HSR action that changes a correct base label to a wrong local label. The records support structured review; they do not guarantee correctness or uniformly beneficial HSR actions.

absolute performance and the paired gap, so preprocessing is part of the scientific comparison rather than an implementation detail.

The methodological contribution is the joint availability of modality-specific labels and margins, traceable fusion weights, a bounded six-label HSR action and cell-level audit records. Branch supervision does not drive the aggregate macro gain in the ablation family; it supplies the inspectable modality interface, while the full locked path is retained for evidence emission. This supports one bounded proposition: for fine-grained RNA-ADT annotation, evidence availability is a deployment resource distinct from aggregate metric optimality. Panel masks, leave-class-out targets and the PDC101 comparison identify concrete failure regimes, including marker-dependent rejection and CD8-boundary sensitivity, rather than establishing universal missing-panel or open-set robustness.

The scope is protocol-defined: the primary inference uses eight PBMC donors, and the 2.47M-parameter capacity control is close but not identical to the 2.51M MOSAIC path. MOSAIC retained a +0.0071 M-F1 difference over that control, so the result supports a joint representation-and-evidence advantage rather than a strict single-module causal claim. The CellTypist row is package-compatible rather than an unqualified native-default reproduction,

and DSB was unavailable without matched background/isotype controls. The representative marker audit covers all 58 labels, while canonical-marker enrichment remains restricted to six sibling labels per modality; attribution cells were sampled from correct predictions, so the analysis is not an error-cell explanation benchmark. The donor-held-out audit score has not been transferred to a secondary cohort with a leakage-safe cell-level artifact. The tagged GitHub package contains the checkpoint, compact evidence tables and CPU audit demo, but no Zenodo DOI or raw-data archive. These boundaries qualify the claim without weakening its practical contribution: MOSAIC makes modality evidence, local action and failure localization inspectable alongside the predicted label.

Data availability

PBMC 3P and 5P data are from GEO accession GSE164378. The COVID-19 cohort is from ArrayExpress E-MTAB-10026. PDC101 is from GEO GSE229791. COMBAT is available through Human Cell Atlas project cdabcf0b-7602-4abf-9afb-3b410e545703, EGA study EGAS00001005493 and Zenodo DOI 10.5281/zenodo.6120249 [10]. Processed split manifests and source-record checks are included with the reproducibility artifacts described in the Supplementary Material.

Code availability

Source code, configuration files, split manifests, compact evidence tables, tests, environment metadata, checksum records, one P1/seed-42 checkpoint and a CPU audit demo are available under the MIT License at <https://github.com/printfnice/MOSAIC> in the tagged submission snapshot. The package includes the audit-record schema and a concrete ten-cell example; raw participant data, cached matrices and full prediction files are not redistributed, and no Zenodo DOI is claimed.

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Author contributions

Hai Yang and Linhao Wu conceived the study. Linhao Wu led the implementation, data processing, experiment execution, result auditing, visualization and manuscript drafting. Hai Yang contributed to method design, experiment planning, interpretation and manuscript drafting. Dan Zhou and Tianyi Zhou contributed to study discussion, biological interpretation and manuscript review. Qin Zhou, Ting Xiao, Qian Zhang and Jing Zhang contributed to data interpretation and manuscript review. Zhe Wang supervised the study and revised the manuscript.

Conflict of interest

The authors declare no competing interests.

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AI-assisted tools were used for code assistance, artifact auditing, table generation and language editing. All analyses, figures, citations

and conclusions were designed, verified and are the responsibility of the human authors.

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